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## CHAPTER 2

### REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter discussed the analysis of the literature related to the study, the systems of various published literature, and the description of the structure of the entire study gathered from books, articles, algorithms or systems (software), and other research materials. This discussion of the manuscript contains a summary of the associated local and foreign literature and systems; it also includes the Synthesis of the Art, Gap Bridged by the Study, and Notes.

#### **Related Literature and Systems**

The following article presents a review of the literature relevant to the present study. It includes foreign and local literature on systems that provide necessary information for a better understanding of the implementation of web and mobile-based profiling systems, the use of RFID technology in attendance monitoring, and data management and security in government systems. This review also discusses the proposed system by the researchers and explains why it is important for them to study and understand these topics.

#### ***Profiling Systems***

The adoption of profiling systems has been instrumental in enhancing data management, record-keeping, and decision-making across various sectors. These systems streamline administrative tasks, improve accessibility, and ensure the accuracy and security of stored information. In the study by Lacasandile et al. [2020], the researchers highlighted the significance of information and communications technology (ICT) in governance. The study emphasized that an automated profiling system could help barangay officials manage household data effectively [22]. The system facilitated budget allocation and decision-making by aggregating essential data,

such as labor statistics, household income, and demographics. The functionality of BIPS was rated highly by users, with a mean score of 4.47 for ease of operation and 4.50 for system reliability, underscoring its efficiency in e-Governance applications.

The "Barangay Profiling System" by Quindatan [2020] further supports the significance of computerized profiling. This system was designed to help barangay employees manage residents' data efficiently, reducing human error and processing time. By digitizing resident records, the system ensured improved accuracy, accessibility, and security, making it a viable alternative to manual data management methods[26]. The researchers concluded that automation enhances organizational efficiency and minimizes the workload of barangay officials. A more specific study by Evangelista et al. [2023] explored the development of a profiling system designed based from the city's needs. The system integrated a comprehensive personal data profile, query mechanisms, and report generation features [14]. Using the XP software development methodology, the system underwent iterative testing phases to ensure its functionality. The study found that the system significantly improved public service delivery by facilitating secure and efficient data management. Meanwhile, the "Barangay Profiling System with Analytics" developed by Jacobe et al. [2021] emphasized the use of statistical analysis in profiling systems. The system allowed authorized users to generate statistical reports based on demographic and socio-economic data [19]. This feature enabled LGUs to make data-driven decisions for community development. The research findings underscored that integrating analytics into profiling systems enhances their effectiveness and usability.

A study by Alinea et al. [2022] examined the role of mobile technology in profiling systems. The research introduced a mobile application that facilitated real-time data entry and retrieval [6]. The system ensured accessibility for barangay personnel. The study concluded that mobile-based profiling systems improve data accuracy, accessibility, and efficiency compared to traditional manual processes. Similarly, Sanchez [2021] conducted an ethnographic study which

focused on demographic, economic, and socio-political aspects of a community. The study employed a descriptive research method to analyze residents' living conditions and access to services [30]. The findings demonstrated that systematic profiling helps local government units (LGUs) design targeted interventions for marginalized communities, reinforcing the need for accurate and comprehensive profiling systems.

Beyond community profiling, a study by Calderon et al. [2020] explored the application of profiling systems in commerce. The system utilized image processing technology to classify customer demographics, providing insights into purchasing behaviors [10]. The study demonstrated that profiling systems could be adapted for various industries to improve decision-making and operational efficiency.

### ***Automated Attendance Monitoring Using RFID Technology***

The integration of Radio Frequency Identification (RFID) technology in attendance and data management systems has been widely explored in various fields, including education and workplace monitoring. RFID-based systems streamline record-keeping, enhance security, and improve efficiency by automating traditionally manual processes. The study by Siddiqui et al. [2024] on the "IoT-Based RFID Attendance System" highlights the importance of real-time attendance tracking using RFID and cloud-based data storage. The system ensures efficient monitoring by automating attendance recording, reducing human intervention, and providing seamless access to records [32]. This study is relevant to the proposed profiling system for Katipunan ng Kabataan (KK) and Sangguniang Kabataan (SK) as it demonstrates the effectiveness of RFID in improving attendance tracking and data retrieval. Similar to the study of P. Kovelan et al. [2020] discusses an IoT-enabled RFID-based system that minimizes manual errors in tracking attendance. It incorporates microcontrollers and GSM/Wi-Fi communication modules to ensure data is stored even during connectivity failures [21]. The redundancy feature of this system aligns

with the goal of the proposed KK and SK profiling system, ensuring continuous and accurate data management even in case of network disruptions.

According to Kaiser et al. [2020], their study titled "Automation of Time and Attendance Using RFID Systems" elaborates on the various applications of RFID for attendance monitoring in universities. It automates student identification, tracks unauthorized entries, and integrates email and SMS notifications [25]. The capability of RFID to automate multiple administrative tasks provides valuable insights for the proposed system, which aims to improve youth profiling and participation monitoring. Previous studies by B.M Sri Madhu et al. [2021] further supports the effectiveness of RFID in attendance management. By integrating IoT with RFID, the system reduces manual effort and improves accessibility through mobile applications [33]. This study reinforces the need for a web and mobile-based platform in the proposed KK and SK profiling system to facilitate remote access and management. Also, the study of Ahmed et al. [2024] highlights RFID's role in workplace efficiency, security, and attendance tracking [2]. The study emphasizes the benefits of cloud-based solutions, which are integral to the proposed system's goal of enhancing transparency and accountability in youth governance.

In the study of Valdez, et al. [2024] titled "TrackID: RFID-Based Student Monitoring System for Accessing Institutional Campus" provides insights into RFID-based student tracking systems, detailing real-time attendance updates and database integration [35]. The findings are relevant in designing an efficient and real-time KK and SK profiling system that ensures secure and accurate record-keeping. This has also been explored in prior studies by Sruthy R et al. [2023] which discusses the security and efficiency aspects of RFID attendance systems. The study suggests that automated attendance systems improve academic and administrative decision-making by providing structured attendance reports [34]. This aligns with the objective of the proposed KK and SK profiling system to enhance efficiency in youth governance.

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***Web and Mobile-Based Systems for Youth Governance***

The adoption of mobile and web-based profiling systems has significantly improved data management, record-keeping, and decision-making across various sectors. These systems streamline administrative tasks, enhance accessibility, and ensure data security. In the study by Algara [2021], a web-based barangay profiling system was developed to manage resident records efficiently. The system enabled barangay personnel to issue certificates, process document requests, and update resident data in real time. By transitioning from manual to digital record-keeping, the system reduced errors and improved accessibility for barangay officials[4]. Moreover, the "Barangay Profiling System" developed by Sarne [2020] further supports the importance of integrating web-based technology in local government operations. This system allowed barangay officials to access and update resident records through a secure online platform. The research emphasized that using a centralized database improves data retrieval, reduces administrative workload, and minimizes record duplication. The study concluded that web-based profiling systems enhance efficiency by providing quick and organized access to community data [26].

A study by Vardoulakis [2020] highlighted the advantages of mobile applications in profiling systems. The research introduced a mobile-based profiling system that enabled barangay personnel to collect and access household records using smartphones and tablets. The system utilized cloud storage for real-time data updates, ensuring that information was accessible anytime and anywhere. The study found that mobile-based systems significantly improved response times for administrative tasks and enhanced the efficiency of barangay data management [36]. Meanwhile, the "Household Profiling System with Analytics" by McFalls [2020] demonstrated the potential of integrating analytics into profiling systems. The system allowed authorized users to generate demographic reports and analyze socio-economic data for community development planning. The research findings emphasized that combining mobile and web-based technologies

with analytics enhances decision-making, resource allocation, and the efficiency of local government services.

A study by Alinea et al. [2022] examined the development of a mobile and web-based household profiling system for Barangay Guinhawa, Quezon. The system was designed to provide barangay personnel with a cross-platform solution that allowed access via both mobile devices and web browsers. The research integrated Node.js and PHP for web-based functionality and Firebase for real-time data synchronization between mobile and web applications. The study found that using a dual-platform system improved data accuracy, accessibility, and efficiency, enabling barangay officials to update and retrieve household information seamlessly [6]. Similarly, Sanchez [2021] conducted a study on the effectiveness of mobile and web-based profiling systems in demographic and socio-economic data collection. The system provided an interactive interface where barangay personnel could input, update, and analyze resident data. The study concluded that web and mobile accessibility enables local government units (LGUs) to provide better services, respond quickly to residents' needs, and develop more accurate community programs.

Beyond community profiling, Calderon et al. [2020] explored the application of mobile and web-based systems in data management and governance. The research introduced a cloud-based profiling system that ensured secure data storage, accessibility, and integration with other government databases. The study demonstrated that mobile and web platforms enhance collaboration, data transparency, and real-time monitoring, making them essential for modern governance and decision-making [20].

The integration of mobile and web-based profiling systems in barangay administration has proven to be a transformative approach to data management. Existing studies highlight their effectiveness in improving service delivery, enhancing accessibility, and ensuring data security. The Mobile and Web-Based Household Profiling System for Barangay Guinhawa builds upon these findings by incorporating real-time database management, cloud storage, and

cross-platform accessibility, ensuring that barangay personnel can efficiently manage household records and provide timely services to residents.

### ***Data Management and Security in Government Systems***

The adoption of digital technologies in government systems highlights the need for secure data management and data privacy, especially in youth governance. This is relevant to our study, as a Profiling System for Katipunan ng Kabataan (KK) and Sangguniang Kabataan (SK) must ensure the confidentiality, integrity, and availability of member data for efficient record-keeping and participation tracking. Implementing robust security measures is essential to protect against unauthorized access, cyber threats, and data breaches, while also addressing regulatory compliance and privacy concerns. Studies on policy-driven security, encryption, and blockchain offer valuable insights into developing a secure and efficient KK and SK profiling system that upholds data protection and governance standards.

The Philippine Data Privacy Act of 2012 (RA 10173) serves as a fundamental law in protecting personal data privacy and security in the Philippines. It mandates both public and private institutions to establish data protection measures and comply with National Privacy Commission (NPC) regulations. In the study of Cheng [2023] emphasized that as digital transformation accelerates in government systems, compliance with the Data Privacy Act has become increasingly critical to prevent data breaches and unauthorized access[11]. Similarly, the research of Pesito et al. [2021] examined the efficiency of Data Privacy Act compliance through a web-based self-survey approach, finding that institutions with structured compliance mechanisms experienced fewer data security risks[9]. Meanwhile, in the study of Pitogo [2021] analyzed government agency compliance, identifying challenges such as policy misalignment and lack of cybersecurity training[31], which could also impact to the KK and SK profiling system if privacy safeguards are not properly implemented. This is highly relevant to the KK and SK Profiling System, which

involves the collection, storage, and processing of youth member data. Ensuring compliance with the Data Privacy Act strengthens data security measures, preventing unauthorized access to sensitive information and maintaining trust and transparency in youth governance. The integration of RFID technology in the system must align with NPC privacy safeguards, ensuring that personal data is handled ethically and securely.

The study of Farid et al. [2023] systematically reviewed digital information security management policies in academic libraries. Their study highlights the role of governmental entities in enforcing security standards for data management and cybersecurity within government institutions[15]. The findings suggest that a structured security management policy significantly improves data protection mechanisms and mitigates potential cyber threat. This is directly applicable to the KK and SK profiling system, that ensuring data privacy and protection is essential when handling youth membership records and participation logs. However, Haraldsdottir and Gunnlaugsdottir [2022] argue that documenting personal information within a regulatory environment is inherently complex[17]. Their study suggests that excessive bureaucratic requirements often hinder efficient information security management instead of improving it. They found that compliance issues and documentation complexity reduce the effectiveness of security policies.

With the rise of smart city technologies, real-time data collection has become essential for efficient governance. However, Ahmad et al. [2022] highlight that without strict access controls, smart city infrastructures are vulnerable to cyber threats and data exploitation[1]. Their study emphasizes the importance of balancing privacy policies with public data accessibility, a challenge that must be addressed in the KK and SK profiling system to ensure data security while allowing efficient tracking of youth participation. Similarly, Chorley [2021] examined security risks in open government data systems, finding that while transparency improves governance, it also increases exposure to cyber threats[12]. The study concludes that without proper encryption and



authentication measures, government databases remain susceptible to breaches. This reinforces the need for strong encryption protocols and secure authentication processes in the KK and SK profiling system to prevent data manipulation and unauthorized access.

In their research, Alharbi et al. [2021] analyzed Saudi Arabia's e-government security framework, emphasizing the effectiveness of encryption techniques and dual-trust authentication models[5]. Their findings suggest that multi-layered security measures significantly reduce vulnerabilities and enhance data protection. However, Giri and Shakya [2021] examined Nepal's e-government security framework, revealing weaknesses in policy implementation and technological infrastructure[16]. Their study found that limited cybersecurity expertise, outdated security measures, and poor risk assessment strategies leave e-government systems vulnerable to hacking attempts and data manipulation.

In accordance with the study discussed Alketbi et al. [2021] explored the potential of blockchain technology in government data security, finding that its decentralized structure enhances transparency, trust, and data integrity[7]. Their study suggests that blockchain mitigates security risks by eliminating centralized vulnerabilities, making government transactions more secure. However, Luthra et al. [2023] highlight that while blockchain offers enhanced data security and transparency, its implementation in government systems faces several challenges, including high costs, complex infrastructure requirements, and integration difficulties with existing databases. The study emphasizes that governments struggle with aligning blockchain with current IT frameworks, making its adoption slow and resource-intensive[23]. This highlights the need for the KK and SK profiling system to evaluate cost-effective security solutions that balance efficiency, affordability, and scalability.

The increasing reliance on digital technologies in government systems highlights the need for robust data management and security frameworks. As explored in various studies, structured security policies, encryption techniques, and emerging technologies such as blockchain play a

crucial role in enhancing data protection and mitigating cyber threats. However, challenges such as regulatory inefficiencies, technological limitations, and implementation costs continue to hinder the effectiveness of these security measures. While governments strive to balance transparency, accessibility, and cybersecurity, it is evident that existing approaches require continuous improvement to address evolving threats. By integrating policy-driven security enforcement, encryption protocols, and advanced data protection mechanisms, the KK and SK profiling system can enhance youth governance, ensure secure record-keeping, and comply with government data security standards. The insights from existing research provide a strong foundation for developing a comprehensive, adaptable, and secure profiling system, ensuring data integrity, privacy protection, and cybersecurity resilience, which are essential for ensuring secure and efficient government systems in the digital era.

### ***System Evaluation Using Black Box Testing***

A critical aspect of system development is black box testing, ensuring the functionality, reliability, and security of the software without examining its internal code structure. Black box testing in RFID-based applications has steadily increased over the years. According to Hasanah et al. [2021], black box testing is crucial in markerless augmented reality (AR) construction engineering, where extreme programming techniques ensure system reliability [18]. The study used Black-Box techniques for Functional Usability testing, highlighting how important RFID and IoT-based technologies require of accurate functional testing to ensure real-world applicability. Similarly, Rahim et al. [2021] examined how black box testing plays a role in automotive IoT applications. The study identified how RFID-enabled black box systems can enhance vehicle tracking and safety[27]. The researchers concluded that black box testing is essential for ensuring the functional correctness of IoT-based real-time tracking applications, which creates a standard for smart governance and security applications. However, Albahli & Andrews

[2021] argued that Model-Based Testing (MBT) is a more effective black box testing methodology compared to traditional equivalence partitioning. Their study on smart home systems using RFID and mobile applications suggests that event-driven finite state machine (EFSM) and controlled event-driven finite state machine (CEFSM) testing improves the detection of state-based errors better than black box testing alone[3].

In the study of Majid et al. [2022], the growth of mobile and web applications integrated with RFID led to an exploration of how black box testing methods can be enhanced with explainable artificial intelligence (XAI)[24]. The study highlighted that traditional black box testing fails to identify security vulnerabilities in mobile-based RFID tracking applications. It suggested incorporating machine learning-based white-box testing to improve coverage and reduce security loopholes. Contrary to Majid et al.'s findings, Rasool et al. [2022] emphasized that black box testing is sufficient for most web and mobile applications, particularly in Internet of Medical Things (IoMT) applications. Their research demonstrated that automated black box testing procedures reduce human error, ensuring faster system acceptance in RFID-based health monitoring systems. However, they acknowledged that black box testing alone cannot detect logic errors[29].

In addition to previous studies, Song et al. [2022] introduced an automated black box testing approach using high-level abstraction, addressing manual labor inefficiencies in functional system testing. Their research demonstrated an automated process that mitigates testing and reviewing issues through a compiler-like automation method, which generates independent test cases and formalized specifications. This process significantly reduces manual effort, improves accuracy, and enhances software reliability in industries like railway signaling and transit systems. Furthermore, it overcomes the common limitations of traditional black box testing, such as repetitive manual work, long test execution times, and specification-related errors, by leveraging an automated testing simulation tool [13].

Furthermore, Anh et al. [2024] conducted a collaborative IoT school bus monitoring system study, integrating RFID tracking and mobile application development. The study compared black box testing with white box testing and found that black box testing is more efficient in testing real-time location tracking. However, they noted black box testing's inability to detect backend algorithmic failures, arguing for a hybrid testing approach [8]. Additionally, Yalli et al. [2024] examined the role of black box testing in IoT-based authentication systems[37]. Their study found that RFID-based security models require more dynamic testing techniques beyond traditional black box testing. While black box testing ensures functional correctness, it fails to uncover security loopholes that may arise due to poor encryption protocols.

Lastly, Rahman et al. [2025] suggested that blockchain integration in RFID applications would require new testing methodologies. Their work proposed a Blockchain-Based IoT testing framework that integrates black box testing with smart contract verification. The study contradicts previous works by suggesting that black box testing alone is inadequate in blockchain-based mobile and web applications[28]. Black box testing remains a vital approach in evaluating RFID-based and IoT applications, ensuring functional correctness and system reliability. However, as highlighted by various studies, its limitations in detecting security vulnerabilities, logic errors, and backend failures necessitate hybrid approaches that integrate Model-Based Testing (MBT), AI-driven automation, and blockchain verification. As technology advances, the future of system testing will rely on automated and security-focused methodologies to address the growing complexity of modern applications, ensuring that RFID-based profiling systems, particularly in government and youth governance, maintain accuracy, security, and compliance with evolving data protection standards. A combination of black box, white-box, and AI-enhanced techniques will be essential in creating more robust, secure, and efficient testing frameworks for government, healthcare, transportation, and smart technology systems.

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## Synthesis of the State-of-the-Art

After reviewing the related literature and existing systems, the significance of profiling systems, web and mobile-based systems for youth governance, RFID technology for attendance monitoring, and secure data management in government operations is evident. The implementation of profiling systems has been widely used in governance, community management, and public service administration. Studies such as those by Lacasandile et al. [2020] and Quindatan [2020] emphasized the benefits of automated profiling in local government units (LGUs) by improving data accuracy, accessibility, and efficiency. Similarly, Evangelista et al. [2023] demonstrated how profiling systems streamline public service delivery through data integration, ensuring efficient record-keeping and decision-making. These studies support the current research's aim to develop a structured profiling system for the Katipunan ng Kabataan (KK) and Sangguniang Kabataan (SK) in Iriga City, facilitating better data retrieval and monitoring youth participation.

The use of RFID for attendance monitoring and security has also been extensively studied. Research such as that of Siddiqui et al. [2024] and Qaiser et al. [2020] highlighted the effectiveness of RFID in automating attendance tracking, minimizing manual errors, and integrating cloud storage for real-time data management. These studies align with the present research's goal of enhancing attendance tracking during SK programs and activities through RFID integration. Furthermore, the study by Ahmed et al. [2024] demonstrated RFID's role in ensuring workplace efficiency and security, reinforcing its application in youth governance for reliable attendance monitoring and access control.

The studies of Evangelista et al. [2023] and Martinez et al. [2022] highlight the role of web-based systems in improving government data management, aligning with the KK and SK profiling system's goal of enhancing youth governance through digital solutions. However, unlike

these studies, which focus on barangay-level resident profiling, this study is specifically designed for youth governance, integrating real-time participation tracking, automated record-keeping, and mobile accessibility to support SK officials in managing youth programs efficiently.

In the study of Santos et al. [2023] and Ramos et al. [2021], the challenges of manual record-keeping in SK governance were emphasized, particularly the difficulty in tracking member participation and generating accurate reports. This study addresses those concerns by developing a web and mobile-based system that allows SK officials to manage youth records, monitor engagement in real time, and generate automated reports, ensuring efficient decision-making and governance transparency. Based on the literature examined, Chua et al. [2022] and Del Rosario [2023] discussed the technical and financial constraints in implementing mobile-based government platforms, particularly in ensuring data synchronization and real-time access across multiple devices. To bridge this gap, this study leverages a cost-effective and scalable web-mobile system that ensures seamless data integration, cloud-based storage, and user-friendly access for SK leaders and KK members.

Additionally, while Gonzalez et al. [2023] and Lim et al. [2022] explored mobile applications for public administration, these systems were not tailored to youth governance and lacked features for program participation tracking and digitalized youth engagement monitoring. In contrast, this study prioritizes a specialized web and mobile-based platform that enhances SK governance by providing real-time data access, streamlined administrative functions, and improved youth participation tracking. By integrating web and mobile technologies, this study presents an innovative and accessible solution for youth governance, ensuring that SK leaders can efficiently manage, monitor, and evaluate youth engagement while addressing the limitations of existing government systems. By analyzing the findings from these studies, the present research integrates key technological advancements and best practices in profiling, attendance tracking, and data

security, ensuring the development of a reliable, secure, and efficient profiling system designed based from the needs of Katipunan ng Kabataan and Sangguniang Kabataan in Iriga City.

### **Gap Bridged by the Study**

The researchers will develop the study by considering the current state of the art, the related literature, and the systems. While similar studies have been conducted, the focus and scope of this research differ from previous studies. Despite the advancements in profiling systems, attendance tracking, and data management, existing studies lack a comprehensive solution developed for youth governance, specifically for the Katipunan ng Kabataan and Sangguniang Kabataan. Previous studies primarily focused on barangay-level profiling systems without integrating attendance tracking and mobile accessibility. The present study addresses this gap by developing a web and mobile-based profiling system with RFID integration, ensuring efficient member registration, attendance monitoring, and secure data management for city-level Sangguniang Kabataan.

While previous research, such as that of Evangelista et al. [2023], explored profiling systems for government use, these systems did not focus on tracking youth participation in governance activities. The proposed system bridges this gap by providing real-time attendance monitoring for SK programs and activities, improving transparency and accountability in youth governance. Furthermore, existing RFID-based attendance systems, as discussed by Siddiqui et al. [2024] and Qaiser et al. [2020], primarily focused on academic and workplace settings. The current research extends RFID applications to the Sangguniang Kabataan sector, ensuring accurate tracking of youth engagement in governance.

Web and mobile-based systems have revolutionized governance by enhancing accessibility, efficiency, and transparency in data management. However, studies by Evangelista et al. [2023] and Martinez et al. [2022] identified limitations in existing barangay profiling systems,

which primarily focus on resident record management and document issuance but lack dedicated functionalities for youth governance. Similarly, Santos et al. [2023] and Ramos et al. [2021] highlighted inefficiencies in manual SK record-keeping, resulting in delays, data inconsistencies, and difficulties in monitoring youth participation. Despite the widespread adoption of e-governance technologies, most systems do not provide real-time accessibility, automated profiling, and centralized data management tailored for the Katipunan ng Kabataan (KK) and Sangguniang Kabataan (SK).

Additionally, Chua et al. [2022] and Del Rosario [2023] emphasized the technical and financial constraints in implementing mobile-based governance solutions, particularly in integrating automated attendance logging, participation tracking, and program evaluation. While studies by Gonzalez et al. [2023] and Lim et al. [2022] explored mobile platforms for public service delivery, these systems were designed for general government operations and lacked youth-specific governance tools that enable SK officials to monitor engagement, generate reports, and manage data seamlessly across multiple devices. To address these gaps, this study introduces a web and mobile-based system for youth governance that streamlines member profiling, program participation tracking, and administrative processes for the Katipunan ng Kabataan and Sangguniang Kabataan. The system allows SK officials to manage youth data, track attendance in activities, and generate reports in real-time through a mobile-responsive interface, ensuring greater accessibility and efficiency. Moreover, automated data synchronization between web and mobile platforms eliminates manual record-keeping errors, while cloud-based storage and encrypted access controls enhance data security and compliance with privacy regulations.

By leveraging web and mobile technologies, this study provides an innovative digital solution for youth governance, optimizing participation tracking, improving transparency, and ensuring data-driven decision-making within the Sangguniang Kabataan leadership structure.



Security remains a crucial aspect of data management in government systems, yet studies by Cheng [2023] and Pitogo [2021] identified compliance challenges with the Philippine Data Privacy Act (RA 10173) in government data handling, including policy misalignment, insufficient cybersecurity training, and inadequate enforcement of privacy safeguards. Similarly, Ahmad et al. [2022] and Chorley [2021] highlighted vulnerabilities in real-time data collection and access control mechanisms, which can lead to unauthorized access, data breaches, and manipulation of government records. Additionally, Farid et al. [2023] and Alharbi et al. [2021] emphasized the need for multi-layered security policies in e-government systems, but existing solutions often lack specialized security frameworks tailored to youth governance and participation tracking. Meanwhile, Luthra et al. [2023] noted that high implementation costs and technical complexities hinder the adoption of advanced security solutions like blockchain in government data management. To address these gaps, this study integrates encryption to ensure the confidentiality of KK and SK member data, multi-factor authentication with OTP verification to prevent unauthorized access, and role-based access control to regulate data access among KK members, SK officers, and Panlungsod na Pederasyon ng mga Sangguniang Kabataan administrators. Additionally, an RFID-enabled attendance tracking system eliminates fraudulent attendance logging, while automated audit logs and secure cloud-based storage enhance data integrity and recovery, ensuring compliance with National Privacy Commission (NPC) regulations through automated data retention policies that protect member privacy. By implementing these security measures, the KK and SK Profiling System strengthens data protection, ensures compliance with privacy laws, prevents unauthorized access, and enhances transparency and accountability in youth governance.

By addressing these gaps, the study significantly contributes to youth governance by developing an efficient and secure profiling system based from the needs of Katipunan ng Kabataan and Sangguniang Kabataan in Iriga City. The integration of RFID technology, mobile accessibility,

and secure data management ensures an advanced solution for streamlined record-keeping, attendance tracking, and efficient governance within the Panlungsod na Pederasyon ng mga Sangguniang Kabataan Office.

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### Notes

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TRACTION

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